

4.3.3 Handling Missing Values (NaN)

خالی خانے NaN ہیں۔

CS XII

Chapter 4: Data and Analysis

Lecture 05 of 8

★ Golden

4.3.3 Handling Missing Values (NaN)

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LEARN IT

Missing cells are **NaN** ("Not a Number"). Statistics crash or lie if you ignore them. Three tools: `isnull()` (True where empty), `fillna(value)` (replace), `dropna()` (delete the row).

```
import pandas as pd
df = pd.DataFrame({
    "Name": ["Ali", "Sara", "Ahmed", "Ayesha"],
    "Marks": [85.0, None, 78.0, 92.0],
    "Grade": ["A", "A", None, "A"],
})
print(df.isnull().sum())
```

OUTPUT

```
Name      0
Marks      1
Grade      1
dtype: int64
```

WORKED EXAMPLE

Replace Marks with the mean of the known marks, and Grade with a label. Known marks 85, 78, 92 → mean = $255/3 = 85.0$.

```
avg = df["Marks"].mean()          # 85.0
df["Marks"] = df["Marks"].fillna(avg)
df["Grade"] = df["Grade"].fillna("Not Assigned")
print(df)
```

OUTPUT

	Name	Marks	Grade
0	Ali	85.0	A
1	Sara	85.0	A
2	Ahmed	78.0	Not Assigned
3	Ayesha	92.0	A

`df = df.dropna()` would instead delete Sara's row and Ahmed's row (any row that still had a NaN).

EXAM TIP

State: identify with `isnull / sum`, fill numeric gaps with the mean, fill categories with a label, or drop rows if filling would invent a lie. Never analyse first and clean second.

COMMON MISTAKE

Filling a salary gap with the mean of a company that includes the CEO can hide poverty (the same outlier problem as the mean). Know *why* the value is missing before you fill.

اردو میں سمجھیں

خالی خانے NaN ہیں۔ `isnull` تلاش، `fillna` بھرنا (اکثر اوسط سے)، `dropna` قطار ہٹانا۔ پہلے صاف کرو، پھر تجزیہ۔

CHECK YOURSELF

1. What does `df.isnull().sum()` return?
2. Does `dropna()` delete a column or a row by default?

Answers: 1. Count of NaNs in each column. 2. Any **row** that contains at least one NaN.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. NaN means:

- a. Not a Number — missing
- b. A huge number
- c. Zero always
- d. A pie slice

MCQ 2. `isnull().sum()` gives:

- a. Count of missing per column
- b. The mean
- c. The PK
- d. The chart type

MCQ 3. `fillna(mean)` :

- a. Replaces gaps with the average
- b. Deletes the column names
- c. Sorts
- d. Commits SQL

MCQ 4. `dropna()` by default:

- a. Drops rows that contain any NaN
- b. Drops the whole DataFrame always
- c. Fills with 0
- d. Draws a box plot

MCQ 5. You should clean NaNs:

- a. Before analysing
- b. After publishing
- c. Never
- d. Only in Figma

MCQ 6. Filling salary with the mean can:

- a. Hide poverty if a CEO is in the data
- b. Fix ethics
- c. Create keys
- d. Route packets

Extra short questions

1. Three tools: find, fill, drop.
2. Mean of 85, None, 78, 92 for filling.
3. When drop is better than fill.
4. `isnull()` returns what type of frame?

Extra long question

1. Given Marks [85, None, 78, 92] and Grade [A,A,None,A]: show `isnull` sum, fill marks with mean and grade with "Not Assigned", print the frame. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. `isnull/sum`; `fillna`; `dropna`.

2. Known $85+78+92=255/3=85.0$.
3. If inventing a value would lie (unknown category, huge outlier).
4. Booleans True where missing.

Long

1. Counts 1 and 1. Sara marks 85.0, Ahmed grade Not Assigned. Code + table. Warn about dropna deleting two rows if used first.

Next lecture: 4.4 Data Organization and Representation